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Intelligent Underwater Mine Detection Using Multi-Backbone Feature Fusion with Temporal Attention-Gated Unit on Side-Scan Sonar Imaging

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Abstract

Underwater mines pose a high risk to submarines and ships. As a result, navies across the globe employ mine countermeasure units to counter them. Among the measurements employed by MCM units is mine hunting, which involves examining every mine in a suspicious region. With the development of sonar image technology, these images are broadly used for oceanographic studies in military and civilian applications. High-resolution image sonars may be mounted on several survey platforms, typically autonomous underwater vehicles that give improved speed and enhanced data quality with longer-range support. The classification and detection stages are generally executed employing a sonar mounted on an underwater vehicle or on a ship's hull. After recovering the sonar data, military personnel scan the seabed imaging for detecting the targets and classifying the benign objects or mine-like objects. In recent years, researchers have examined the application of deep learning for the classification and detection of mine-like objects from sonar images. This manuscript focuses on the design of Intelligent Mine Detection and Object Recognition Using Multi-Backbone

Feature Fusion (IMDOR-MBFF) approach in Side-Scan Sonar Imaging data. The main objective of the IMDOR-MBFF framework is to effectively improve detection and classification performance for real-time underwater mine detection using autonomous underwater vehicles. Primarily, a gamma distribution-based speckle noise model with logarithmic transformation was deployed as the image pre-processing phase to suppress multiplicative noise and enhance structural details. Subsequently, object detection is performed using a RetinaNet architecture with backbone fusion to accurately localize mine-like targets under complex seabed conditions. In addition, discriminative features are extracted through a lightweight yet powerful fusion of GhostNetV2, ConvNeXt, and ShuffleNetV1 is capturing both fine-grained textures and global contextual information. After feature extractor, a feature fusion strategy based on concatenation has been employed to efficiently integrate the complementary information extracted from the three backbones. Finally, a temporal attention-gated unit was employed for accurate mine classification by modeling temporal dependencies across sonar image sequences. The experimental results of the IMDOR-MBFF framework are evaluated on benchmark datasets, and the comparative analysis demonstrates its superior performance over existing methodologies across multiple evaluation metrics.

Keywords: Mine Detection; Object Detection; Side-Scan Sonar Imaging; Mine-Like Objects; Deep Learning; Feature Fusion

1. Introduction

Side-scan sonar image analysis in the domain of mine countermeasures (MCM) is usually carried out by an expert [1]. This examination is complicated because of the huge variability in the presence of the side-scan images, along with the higher levels of noise that normally occur in the images. In the last twenty years, autonomous underwater vehicles (AUVs) have been progressively utilized to monitor the Underground Ocean [2].

AUVs provide a capable platform for increasing higher-resolution imaging sonars, such as synthetic aperture or side-scan sonars. Related to lidars and radars, sonars are compatible for detecting small objects projected from the underwater because of their capabilities to imagine the lively underwater ecosystems [3]. Sound waves are more broadly propagated than those of light waves and electromagnetic waves, because of their dispersion and low reduction in water [4]. Related to sonars, visual sensors are a more efficient detection method for water-driven actions in lower-perceptibility conditions, such as turbid conditions or low-light.

Automatic detection of mine-like objects (MLOs) in sonar images is a significant task for a mine clearance system, fascinating significant study attention [5]. As an economical technique in unequal warfare, underwater mines are usually utilized to avoid seaways and block marine activities. Also, underwater mines may cause continuous environmental degradation because of the hazardous explosives [6]. In spite of its higher demand in mine detection, the emergence of an automatic method for MLO identification is stimulating for many factors. Initially, an adequate number of labeled data is essential to train a detection system [7]. But, mine models are enormously restricted compared to other object detection tasks due to the expensive and laborious data collection [8]. Next, the acoustic features of echoes differ considerably depending on the extent and perspective of sound pulses. Consequently, an MLO (containing its shadow) is frequently imaged with numerous forms that lead to complications for the detection operation [9]. Sonar images naturally contain the reverberation produced when transferred acoustic beams strike the limits, such as underwater and the water surface. The reverberation causes severe difficulties, particularly in shallow water. Subsequently, the clutter can control the related and cover the overall objective items [10].

Artificial intelligence (AI) is developed for increasing the capability and performance of AUVs. The capability of AI is to identify patterns, huge data

sets, and make informed estimates can expressively increase AUV navigation, collision prevention, and real-life decision-making [11]. In the last ten years, Deep learning (DL) and machine learning (ML) methods were implemented to increase feature detection in collected data by AUVs, namely shipwrecks or detecting mines. DL provides an efficient solution to underwater object detection and tracking challenges [12]. Especially, Convolutional neural network (CNN) is broadly utilized for pattern detection in tasks relevant to computer vision. CNN-driven techniques have attained extensive employment in the object detection process, attaining expressively enhanced detection operation [13]. In a more difficult underwater acoustic landscape and multiple-class object detection context, these techniques display faults like poor real-world performance and inadequate robustness.

1.1. Research Contributions

This manuscript focuses on the design of an Intelligent Mine Detection and Object Recognition Using Multi-Backbone Feature Fusion (IMDOR-MBFF) approach in Side-Scan Sonar Imaging data. The proposed method introduces a novel multi-backbone feature fusion strategy that combines the strengths of GhostNetV2, ConvNeXt, and ShuffleNetV1 to capture both fine-grained and globally related features from side-scan sonar images. Unlike conventional single-model algorithms, the model leverages balancing illustrations to complete an effectual concatenation-based fusion mechanism. Besides, a TAGU is integrated to method sequential dependencies in sonar data, improving detection accuracy. This combined design delivers a lightweight yet robust solution for intelligent underwater mine detection. The main objective of the IMDOR-MBFF framework is to effectively improve detection performance for real-time underwater mine detection using autonomous underwater vehicles. The framework incorporates gamma-based speckle noise modeling, RetinaNet-based object detection, multi-backbone feature fusion using GhostNetV2, ConvNeXt, and ShuffleNetV1, and classification. Experimental outcomes on benchmark datasets exhibits

superior performance compared to existing approaches. The major contributions of this paper are outlined as follows:

- A gamma distribution-based speckle noise model with logarithmic transformation was exploited to effectively suppress multiplicative noise and improve sonar imaging quality.
- A RetinaNet-based object detection architecture with multi-backbone fusion models including GhostNetV2, ConvNeXt, and ShuffleNetV1 was utilized to accurately localize mine-like objects under complex underwater conditions, while a concatenation-based feature fusion strategy efficiently integrates complementary deep representations.
- A temporal attention-gated unit (TAGU) method has incorporated to model temporal dependencies across sonar image sequences for accurate mine classification.
- Extensive experimental validation on side-scan sonar imaging for mine detection dataset demonstrates the superiority of the proposed framework over existing methods across multiple evaluation metrics.

1.2. Paper Organization

The subsequent sections of this paper are structured as follows:

- Section 2 offers an overview of earlier methodologies and approaches to underwater mine detection using sonar imaging data.
- Section 3 delineates all proposed methodologies, including preprocessing, object detection, feature fusion, and classification, along with performance evaluation.
- Section 4 describes and analyzes the results obtained from the proposed methodologies, including comparative analysis.
- Section 5 presents the conclusions and summarizes the general findings of this study.

2. Recent Studies on Underwater Mine Detection Approaches

This section offers a comprehensive review of existing literature on underwater mine detection using sonar imaging data. Su et al. [14] presented a higher-accuracy underwater object detection framework for SAS imagery, known as the HAUOD method. Initially, considering SAS image features, a sonar pre-processing method is intended to improve the signal-to-noise ratio of object characteristics. An innovative C2fD technique substitutes the unique C2f technique to support observation abilities for edge-blurred regions and lower-contrast objects. Wang et al. [15] presented an innovative multi-level feature aggregation network (MLFANet). Particularly, reduce the effect of underwater reverberation noise, design a LFAM that increases key lower-level image characteristics, like edges, textures, and contours in the object areas. Provide the usual existence of shadow interfering in sonar image, the discriminative feature extraction module (DFEM) to overpower redundant features in the shadow areas, and highlight the object feature regions. Cai et al. [16] presented a multi-modality object identification algorithm depend on underwater camera and sonar. Particularly, for zero-shot object detection, using shadow mapping depends on the imaging principle of sonar to evaluate the characteristics of shadow objects in the domain of sonar images.

Lu et al. [17] proposed an AquaYOLO and developed YOLOv8 version specially designed to increase the accuracy of object identification in subsea sonar imageries. AquaYOLO substitutes standard convolution layers with remaining blocks in the backbone network by improving the feature extraction. Also, Context-Aware Feature Selection (CAFS) and Dynamic Selection Aggregation Module (DSAM) in the neck system. Qin et al. [18] recommended an object detection technique known as YOLOv7C that depends on deep learning (DL) to overcome these issues. A consideration mechanism is combined with the support network of the method to increase its capability to manage complicated backgrounds in sonar images and efficiently extract features. Current redundancy pruning has been utilized for

substantially decrease the size of model during handling advanced detection results. Zhao et al. [19] presented a new transfer learning method using a customized You Only Look Once (YOLO) system, underwater sonar (UWS)-YOLO. This method alleviates overfitting in small data by transmitting features from underwater optical images to sonar images. UWS-YOLO is intended to handle blurred small targets within lower-resolution sonar images, integrating an orthogonal network attention mechanism to augment attention to subtle details and a flexible serpent convolution method to increase recognition of tubular structures.

Huang et al. [20] presented a Seg2Sonar framework that depends on spatially adaptive denormalization (SPADE). This framework produces images with separation maps, consequently removing the need for sample annotation. Moreover, to combine the SLE method into the SPADE model to improve feature extraction capability with minimal data. By increasing the practicality of produced images, the FFL method and the EL technique were develop the random integration capability of the network, considering the features of severe distortion and low resolution of sonar images. Zhou et al. [21] presented an innovative style transfer method, such as multigranular feature alignment cycle-consistent generative adversarial network (CycleGAN), to produce sonar images utilizing remote sensing images that can mitigate the dependency on original sonar data. Particularly, add a spatial attention-driven feature combination technique to preserve unique features by attending to illustration parts of an image. A couple of cross-domain discriminators are intended to direct generators to produce images that take on sonar styles. Lu et al. [22] present Dynamic Selection Aggregation Module (DSAM) and Context-Aware Feature Selection (CAFS) in the neck network. These changes let AquaYOLO to capture complicated particulars improved and decrease feature idleness, important to better performance in underwater object detection tasks. Table 1 presents existing methods for underwater mine detection, including techniques, datasets, and key findings.

Table 1 Comparative analysis of underwater mine detection models

Authors	Year	Research aims	Techniques	Used datasets	Key findings
Su et al. [14]	2025	To develop a high-accuracy and computationally efficient underwater object detection (OD) in SAS images.	YOLOv8 model	SCTD dataset.	mAP of 95.1%.
Wang et al. [15]	2025	To enhance accuracy and robustness of forward-looking sonar image OD in complex underwater environments.	MLFANet model	UATD dataset.	mAP of 81.86%.
Cai et al. [16]	2025	To enable accurate underwater OD through sonar-camera fusion while reducing data collection and labeling costs.	Supervised learning model	Underwater sonar image dataset.	Accuracy of 81.6%.
Lu et al. [17]	2025	To improve underwater sonar image OD accuracy by enhancing feature extraction and selection within	YOLOv8 models	UATD and Marine Debris datasets.	mAP50 of 94.9% and 97.6%.

		a YOLO-based framework.			
Qin et al. [18]	2024	To develop the accuracy and speed of OD in sonar images under challenging underwater conditions.	YOLOv7 algorithm	Underwater sonar image dataset.	Achieves better detection.
Zhao et al. [19]	2024	To enable sonar-based underwater target detection accuracy when training data are limited.	Transfer learning method	UATD and RUOD datasets.	mAP@0.5 of 3.5%.
Huang et al. [20]	2024	To generate diverse and realistic sonar images for underwater targets, enabling effective detection and recognition with very few or no real samples.	DL model	UATD dataset.	Accuracy of 90%.
Zhou et al. [21]	2024	To create high-quality synthetic sonar images to improve OD performance when real sonar data is limited.	GAN model	Underwater sonar image dataset.	Higher recognition results.

Lu et al. [22]	2025	To improve underwater sonar object detection the accuracy using a enhanced YOLOv8 method with progressive feature extraction and aggregation	AquaYOLO	UATD and Marine Debris dataset	mAP50 of UATD: 94.90% Marine Debris: 97.60%
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3. Methodological Framework

This work proposes an IMDOR-MBFF framework for underwater mine detection utilizing side-scan sonar imagery. This study introduces a stepwise intelligent framework for underwater mine detection and object recognition. Fig. 1 represents the entire workflow of the proposed IMDOR-MBFF algorithm. First, a gamma-based speckle noise modeling approach with logarithmic transformation is applied to suppress multiplicative noise and enhance structural information. Second, mine-like targets are localized using a RetinaNet-based object detection module. Third, discriminative features are extracted through a fusion of GhostNetV2, ConvNeXt, and ShuffleNetV1 to capture multi-scale spatial and contextual information. Fourth, a concatenation-based fusion strategy integrates the extracted features into a unified representation. Finally, a TAGU model temporal dependencies across sonar image sequences to achieve accurate mine classification.

3.1. Preprocessing Technique

Primarily, a gamma distribution-based speckle noise modeling technique was executed to improve image quality. The relative intricacy of noise source in sonar imagery making it essential for examining the patterns [23]. Primarily,

it aids for denoising the features of noise, which assists in yielding a pretend sonar data that is essential because it is challenging for SSS alone to gather an adequate count of equivalent pairs of images comprising noisy imagery. In reality, an image comprises numerous kinds of noise.

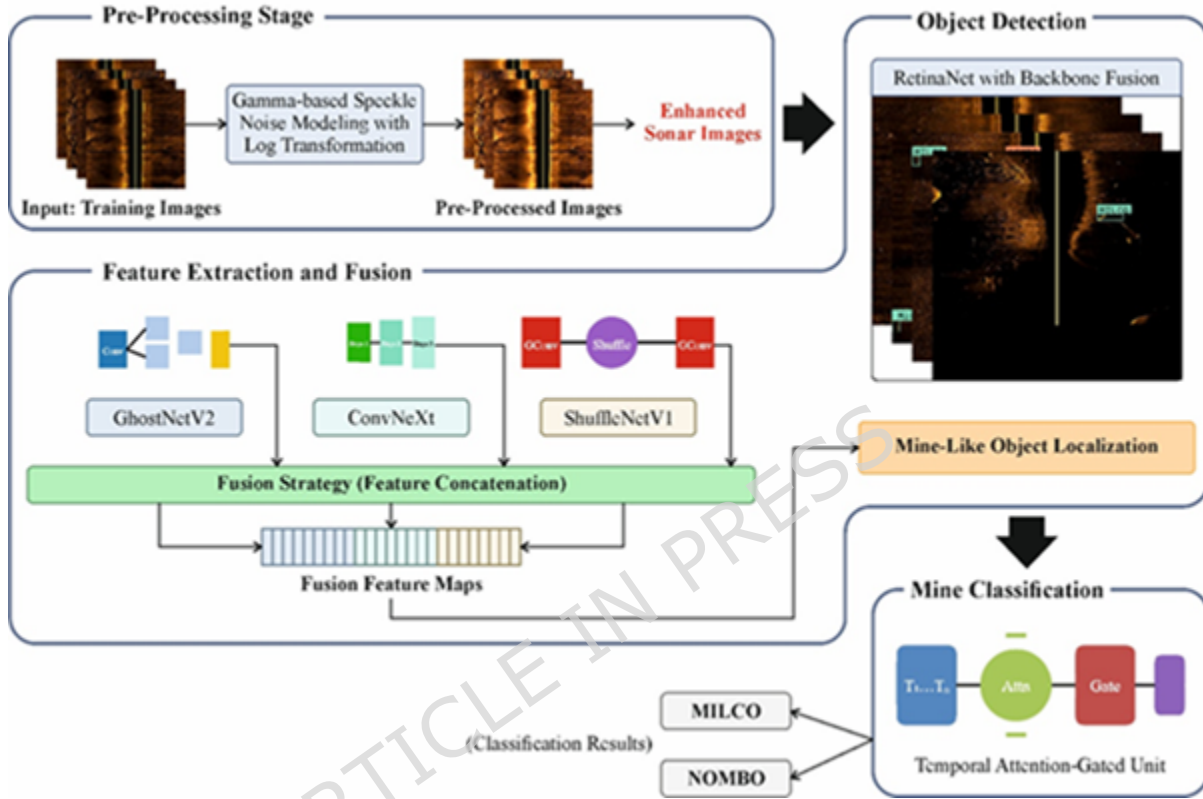


Fig. 1. Overall flow of IMDOR-MBFF approach

According to the relations among noise and echo signal, sonar noise comprises dual major kinds: additive noise initiating the external setting, and multiplicative noise initiated from sonar echoes.

$$y = xn + a \quad (1)$$

Now, w and h implies width and height of the image, correspondingly; $y \in \mathbb{R}^{w \times h}$ denotes observed image which is corrupted by noise; $a \in \mathbb{R}^{w \times h}$ signifies additive noise; $n \in \mathbb{R}^{w \times h}$ refers to multiplicative noise, and $x \in \mathbb{R}^{w \times h}$ implies a probable clean image.

Speckle noise is a multiplicative noise, which means that the intensity of noise is connected with the intensity of sonar signal.

$$y = xn \quad (2)$$

Now, the speckle noise n formed by the Gamma distribution, its probability density function, has been given.

$$f(n) = \frac{n^{\alpha-1}}{\beta^\alpha \Gamma(\alpha)} e^{-\frac{n}{\beta}}, n > 0 \quad (3)$$

However, β denotes parameter of scale, α implies parameter of shape, and $\Gamma(\cdot)$ refers to the Gamma function. Alternative values of α and β are equivalent to speckle noise of diverse levels.

$$E(n) = \alpha\beta \quad (4)$$

$$\text{Var}(n) = \alpha\beta^2 \quad (5)$$

Here, the mean of speckle noise was set to 1 and the variance to $1/L$.

$$f(n) = \frac{L^L n^{L-1}}{\Gamma(L)} e^{-Ln}, n > 0 \quad (6)$$

The logarithm of both sides in the multiplication model has been modified into an additive one.

$$\ln y = \ln (xn) \quad (7)$$

$$\ln y = \ln x + \ln n \quad (8)$$

$$\tilde{y} = \tilde{x} + \tilde{n} \quad (9)$$

Here $\tilde{y} = \ln y$, $\tilde{x} = \ln x$, and $\tilde{n} = \ln n$.

Primarily, an intricate random field of Gamma distribution has been created. Modify the value of variance $1/L$, and diverse levels of speckle noise were attained. Subsequently, the attained speckling noise mapping and clean

image were acquired as logarithms. Because the mean value of Gamma distribution is set to one, the random noise gained once acquiring the logarithm is both negative and positive.

3.2. Object Detection using RetinaNet

Mine localization is achieved through a RetinaNet-based object detection approach. The RetinaNet module is the basic element that performs the fundamental task of eliminating deeper aspects from the input image and carries out object detection (OD) [24]. This method leverages an effectual single-stage framework to accomplish a balance among higher outcomes and real-time execution in the task of OD, and is well-suited for OD in intricate scenes.

The backbone network accepts RetinaNet as the feature extractor, functioning to acquire multi-stage aspects from input imageries, enclose higher-level semantical data at lower resolution, and spatial data at higher resolution. RetinaNet effectually process inputting images that yield multi-scale mapping features, the foundation of succeeding feature fusion and improvement. Subsequently, these multi-scale mapping features were passed to FPN for succeeding multi-level data incorporation. FPN utilizes top-down feature propagation and lateral associations by integrating higher-level semantic aspects with lower-level spatial data, providing the recognition method with sturdy multi-scale perception. The incorporation of multi-layer aspects enhances smaller OD outcomes and upgrades the resilience of the method to objects of differing scales.

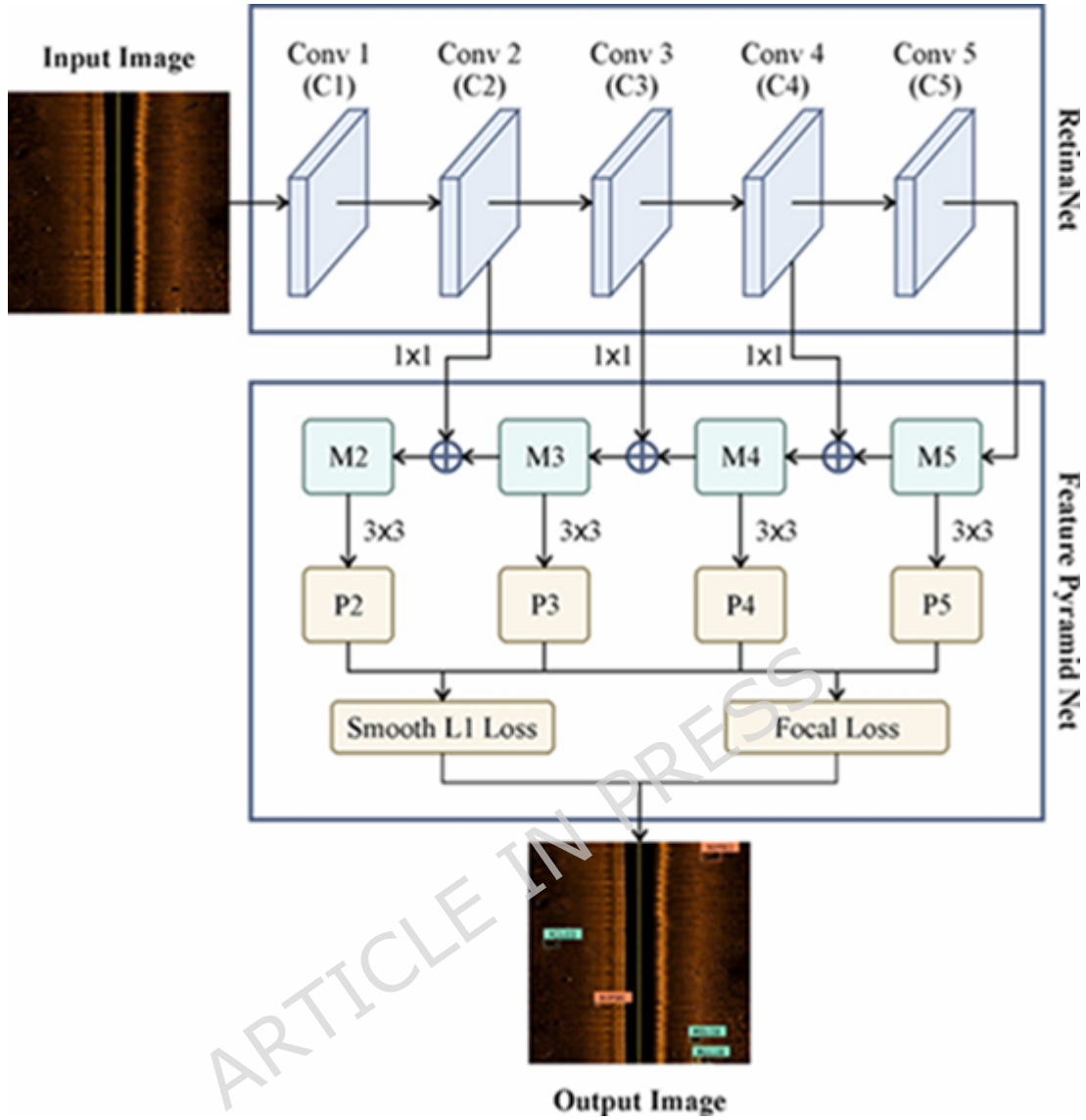


Fig. 2. Architecture of RetinaNet model

Afterwards, the image of input was managed by the network's backbone; the primary function of RetinaNet that yields candidates from these mapping features. These areas were usually created by utilizing a set of pre-defined anchor boxes, equivalent to diverse scales and feature ratios. Assume the predicted class for anchor is p_t , the classification loss is specified:

$$L_{cls} = - \sum_{i \in C} p_t \log(\hat{p}_i) \quad (10)$$

Now, C indicates the set of each probable class and $\hat{p}_i$ implies the predicted likelihood for the class.

$$L_{\text{local}} = -\alpha_t(1 - p_t)^\gamma \log(p_t) \quad (11)$$

Now, α_t signifies balance factor, γ denotes modulation factor, and p_t implies the projected probability for the appropriate class.

The regression branch enhances the targeted position prediction by smoothen L1-loss. With this approach, RetinaNet can precisely categorize and detect every candidate box, specifically for scenes with targets of numerous scales.

$$L_{\text{box}} = \sum_{i \in \{x, y, w, h\}} \left(\frac{1}{N} \sum_{j=1}^N (t_j - p_j)^2 \right) \quad (12)$$

Now, p_j signifies the predicted co-ordinates, N implies regression dimension, and t_j refers to true bounding box co-ordinates.

The RetinaNet model is a fundamental target recognition module that can effectively carry out feature extraction and targeted position, and offers a solid foundation for succeeding feature fusion and improvement. Fig. 2 represents the architecture of RetinaNet model.

3.3. Deep Feature Fusion Network

In this section, deep features are extracted and fused using GhostNetV2, ConvNeXt, and ShuffleNetV1 models.

3.3.1. GhostNetV2 Model

The structure of GhostNet-V2 network is a lightweight CNN [25]. GhostNet-V2 maintains the lower-cost and lightweight linear operation of preceding variant of GhostNet. GhostNet-V2 employs dual kinds of bottleneck components, the primary kind is GhostNet-V2 bottleneck module with a

single size of step, and the improved aspects are input to the secondary Ghost module after the merging of dot product, then input and output were linked with Shortcut, and eventually the features are outcomes. The secondary stage of GhostNetV2 bottleneck module contains dual stages with depth separable convolutional module after the merging of dot product, downsampling the mapping feature, and passes the final Ghost module by accomplishing the amalgamation of channel counts.

$$X_o = X * F_{1 \times 1} \quad (13)$$

However, $F_{1 \times 1}$ signifies convolutional kernel; $*$ implies convolution operation; $X_o \in \square^{H \times W \times C_o}$ refers to outcome features $C_i > C_o$, which means the counts of input channels are superior to the counts of outcome channels.

The secondary stage is to carry out simple linear operations on mapping features in the primary stage to yield similar mapping features, and eventually stack all the mapping features.

$$Y = \text{Concat}([X_o, X_o * F_{DWC}]) \quad (14)$$

Now, $Y \in \square^{H \times W \times C_o}$ implies the outcome features and F_{DWC} signifies the depth of detachable convolutional kernels.

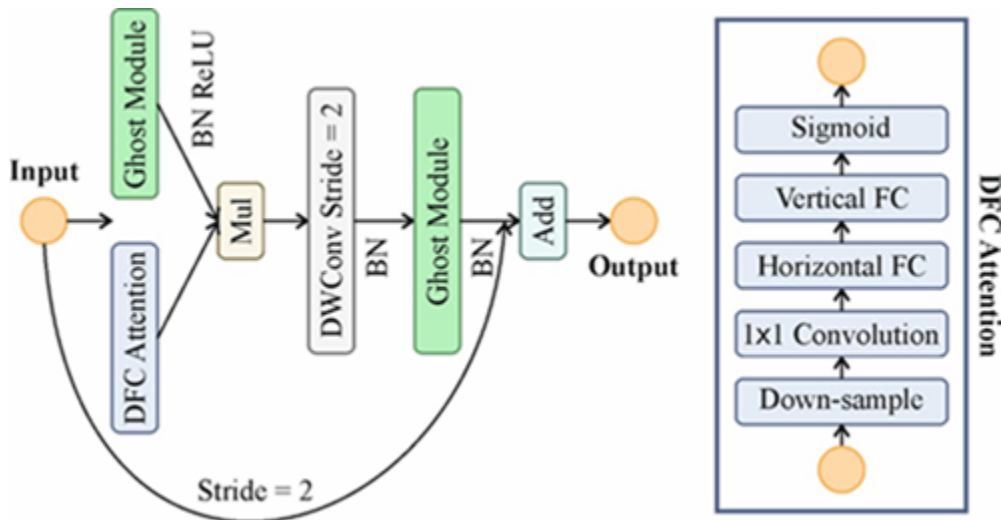


Fig. 3. Structure of GhostNet-V2 model

This approach could effectively lessen the computation cost of method and diminish the number of networks by utilizing ordinary convolution to attain the effective layers of features. Fig. 3 depicts the infrastructure of GhostNet-V2 network.

3.3.2. ConvNeXt Method

The ConvNext is a CNN-based method that aims to enlarge the presentation of techniques for visual tasks, encompassing the diversities that were created by the Transformer [26]. ConvNext enhances the proficiency of the method for classifying images, with an ideal focus on categorization of background data. Thus, ConvNeXt exhibits greater outcomes in the task of image classification.

Depthwise Conv2d processes every input channel individually, instead of mixing every channel conjointly like regular Conv2d. The objective is to lessen the counts of parameter and computation complexity, while acquiring the spatial patterns of every feature channel for higher efficacy.

$$Y_{i,j,c} = \sum_{m=0}^{k_h-1} \sum_{n=0}^{k_w-1} W_{m,n,c} \cdot X_{i+m,j+n,c} \quad (15)$$

Layer Normalization (LN) standardizes activation values in feature dimensions by stabilizing data distribution. This aids in increasing convergence in training, lessens dependence on weight initialization, and upholds stability of gradient flow, making the method simpler to train.

$$\hat{x} = \frac{x_i - \mu}{\sqrt{\sigma^2 + \epsilon}} \quad (16)$$

Conv2d is employed at this phase to present the depthwise conv feature outcomes into a higher-dimension space. This operation integrates data

among channels, thus improving the representation of features that are more intricate than spatial per channel.

$$Y_{i,j,f} = \sum_{c=1}^{C_{in}} W_{c,f} \cdot X_{i,j,c} + b_f \quad (17)$$

Gaussian Error Linear Unit (GELU) is a non-linear activation function that is smoother than ReLU. GELU conceals negative values with probability, instead of cutting them off instantly, thus assisting techniques to learn richer depictions and increasing outcomes in numerous vision tasks.

$$\text{GELU}(x) = 0.5 \cdot x \cdot (\tanh[\sqrt{\frac{2}{\pi}}(x + 0.044715x^3)] + 1) \quad (18)$$

The second Conv2d helps to restore the feature dimension to its original form after it is extended. In this method, the block could acquire intricate non-linear depictions in high-dimensional space, while remaining effective by restoring the size of the output to match the input.

$$Z_{i,j,f} = \sum_{c=1}^{4-C_{in}} W'_{c,f} \cdot Y_{i,j,c} + b'_f \quad (19)$$

Layer Scale is a trainable scalar parameter employed for balancing the outcome contribution of blocks. Primarily, it is initialized with a smaller value by making the network steadier at the start of training, and then its value could progress based on the requirements of the methods.

$$Y = X + \gamma \cdot F(X) \quad (20)$$

Drop Path is a form of regularization in the form of stochastic depth, where specific residual paths were arbitrarily “removed” in training. This model hinders overfitting, increases generalization, and forms a steadier model by training numerous alternative data paths.

$$\text{Drop Path}(x) = \begin{cases} 0, & \text{with probability } p \\ x, & \text{with probability } 1 - p \end{cases} \quad (21)$$

3.3.3. ShuffleNetV1 Model

ShuffleNet-V1 has an innovative CNN framework with exceedingly great computation effectiveness [27]. The main principle includes presenting new dual operations like channel shuffling and pointwise group convolution (gc) that can considerably decrease computational complexity, while preserving model accuracy. Firstly, the Pointwise gc minimizes the computation complexity of 1×1 convolutions by using gc in the 1×1 layer, confirming every convolution process only on relevant input channel collections, thus greatly decreasing the expenses. Secondly, Channel shuffle overcomes the insufficiency of collective data exchange in gc by consistently shuffling channel data from input mapping features. Particularly, intended for a convolutional layer with g groups and $g \times n$ output networks, the output channel size is reformed to (g, n) , transferred, and compressed as inputs for the following layers. Such functions give input and output channels that have been completely connected.

Consider an input mapping features with output channels C_o , height H , width W , channel count C , and kernel size $K \times K$. The number of factors for all the convolution types has been measured as given by

- standard convolution (sconv) variables: $P_{sc} = C_i \times C_o \times K \times K$;
- group convolution (gc) parameters: $P_{gc} = \frac{1}{g} \times C_o \times K \times K$;
- depth-wise separate convolution (dw) factors: $P_{dw} = C_i \times K \times K$.

Therefore, sconf is more higher number of parameters, gc represented as $1/g$, and dw holds a lesser number, at $1/C_o$ of standards. Integrating gc and dw significantly diminishes network computations and parameters.

The ShuffleNet module is formed based on bottleneck component values. At its residual branch, an efficient 3×3 is utilized for the 3×3 layer. The primary 1×1 layer has been exchanged with pointwise gc accompanied by channel

shuffles, and secondary pointwise gc reset channel sizes for matching the alternate way. After the step is used, a 3×3 average pooling has been included in the shortcuts pathway, and element-wise calculation is changed with channel connection. ShuffleNet V1 comprises stacked ShuffleNet modules separated into phases, with the primary structure block in every phase relating stride = 2, keeping alternative hyperparameters in the phases, and expanding output channels for the following phase. Insufficiency and complexity have been measured by changing channel scaling parameter s and group number g for achieving lightweight feature extraction.

3.3.4. Feature Fusion Strategy

Next to the feature extraction, a feature fusion approach has been used for efficiently combining the many extracted data with a 3 backbone architectures like ShuffleNetV1, ConvNeXt, and GhostNetV2 [28]. Every backbone gains various features of the sonar images, namely multi-modal data, architectural data, and texture patterns, confirming a higher representation of prospective underwater mines. The features are derived from the above-mentioned three architecture can be integrated by employing a channel-wise concatenation process, establishing an integrated feature matrix as given in Eq. (11):

$$F_{\text{fused}} = \text{concat}(F_G, F_C, F_S) \quad (22)$$

Here, $F_{\text{fused}} \in \mathbb{R}^{(D_G+D_C+D_S) \times W}$ signifies the fused feature vector, $F_G \in \mathbb{R}^{D_G \times W}$, $F_C \in \mathbb{R}^{D_C \times W}$, and $F_S \in \mathbb{R}^{D_S \times W}$ refers to the feature maps removed from ShuffleNetV1, ConvNeXt, and GhostNetV2, correspondingly. Next, D_G, D_C, D_S indicate the feature sizes of every backbone, and W represents the spatial width or sliding window of the feature maps. This fusion technique keeps the additional data using a 3 backbones, improving the discriminatory ability of the model by collecting various sonar image patterns.

3.4. Temporal Attention-Gated Unit-Based Classification

Finally, TAGU-based classification is employed for accurate mine recognition. The TAGU algorithm is developed to perform significant tasks in modeling long-term temporal dependency for classifying objects [29]. The TAGU, even when useful to static sonar images, supports highlighting the helpful spatial features by adaptively weighting feature reactions. It improves robustness to noise and eliminate noise exist in sonar data, which develops discernment between background and mine-like objects. Furthermore, it allows the method to capture implied contextual dependences, leading to more accurate classification. Conventional temporal models like traditional GRU normally face the challenge of efficiently gaining the long-term and short-term temporal dependencies, faces restrictions like representation flexibility and vanishing gradients. In order to address these boundaries, TAGU newly incorporates GRU and self-attention within a combined temporal model.

Especially, at every time step t , TAGU initially utilizes GRU-based gating to gain a local temporal model proficiently. The update gate, reset gate r_t , and participant hidden state $\tilde{h}_t$ have been calculated as given by,

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t] + b_z), \quad (23)$$

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t] + b_r), \quad (24)$$

$$\tilde{h}_t = \tanh(W_h \cdot [r_t \odot h_{t-1}, x_t] + b_h), \quad (25)$$

Now, $h_{t-1} \in \mathbb{R}^H$ denotes the hidden state after the prior time-step W_z, W_r, W_h , $x_t \in \mathbb{R}^F$ signifies the input feature vector during series, and b_z, b_r, b_h describes learning factors, σ represents the sigmoid activation function. The ultimate hidden state during the time t can be restructured through the conventional GRU gating approach:

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t. \quad (26)$$

In order to clearly overcome the intrinsic data of successive models, specifically the loss of significant historical data because of compression into

an invariable-size hidden state, TAGU presents a multi-head self-attention mechanism that can be directly applied to the hidden states generated through the GRU. Correctly, the self-attention mechanism calculates contextually enhanced temporal representations H_{attn} as given by:

$$H_{\text{attn}} = \text{Softmax} \left(\frac{Q \cdot K^T}{\sqrt{d_k}} \right) V, \quad (27)$$

where the d_k indicates the dimensionality of main vectors, value (V), key (K), and query (Q) matrices have been estimated for GRU hidden states. Different standard GRUs, this pattern permits TAGU to adaptively assist through every previous hidden state, allocating greater weights to time steps with superior implication, thus efficiently maintaining major historical data with respect to long extended temporal varieties.

The major advancement of TAGU exists in the deep combination of GRU-based gating along with self-attention. This incorporation is not a modest stacking of temporal mechanisms; however, a wisely developed distributed framework, considerably improving the ability of model to obtain temporal dependency in complex information. Precisely, TAGU allows actual gain and analysis of prominent temporal patterns, making it individually appropriate for real-world intellectual object detection environments that are required the both higher computational effectiveness and interpretability. Fig. 4 exemplifies the structure of the TAGU, which includes backward and forward feature transmission with an attention gating mechanism. This gating procedure improves feature refinement and develops contextual representation for accurate classification.

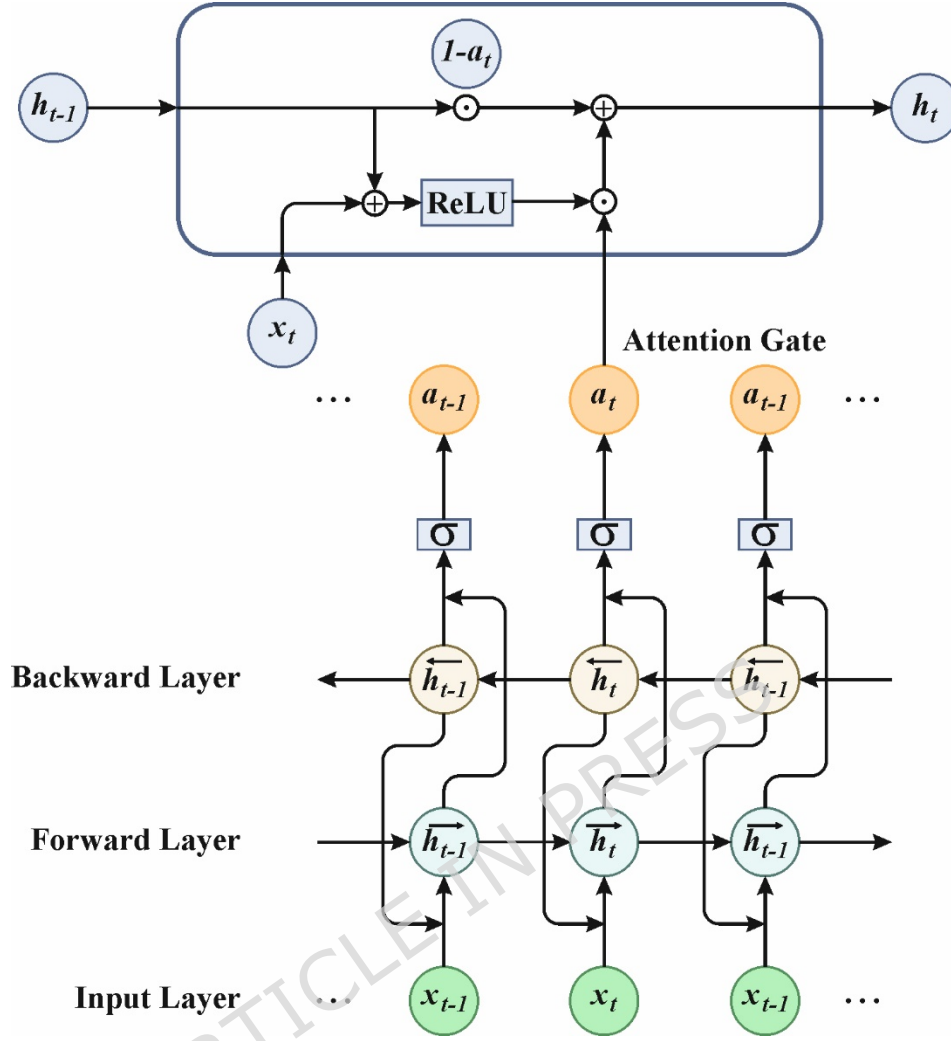


Fig. 4. Structure of TAGU

3.5. Evaluation Metrics

To assess the execution of classification technique, conventional evaluation metrics were utilized.

Accuracy: Determine the total ratio of properly categorized instances. It is described by:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (28)$$

Here, TN and TP indicate true negatives and true positives; FN and FP imply false negatives and false positives, correspondingly.

Precision: Assesses the quantity of appropriately predicted positive samples between each and every positive outcome. It can be computed as:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (29)$$

Precision is a greater precision value that verifies fewer false positives, decreasing inessential measures.

Recall: The capability of the method to detect definite positive cases.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (30)$$

Higher recall confirms that the method does not error a substantial number of instances.

F1-Score: Gives a balanced amount of recall and precision:

$$\text{F1 - Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (31)$$

MCC: MCC is an effective measurement for assessing classification complexities. The value ranges are between -1 and 1 , whereas 1 is accurate prediction, 0 means arbitrary prediction, and -1 representing whole mis-prediction. The computational formula of MCC is represented by:

$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (32)$$

IoU: IoU is a metric that measures the similarity among a predicted and ground truth bounding box. It is determined as the area of their intersection separated by the area of their union.

$$IoU = \frac{\text{Area of Intersection}}{\text{Area of Union}} \quad (33)$$

mAP: mAP is an individual metric that estimates the general performance of an object detection system over all classes. This computation encompasses numerous stages that are comprised of utilizing IoU for determining when a detection is precise. The overall mathematical formula for mAP remains the average of the Average Precision (AP) rates for every class:

$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i \quad (34)$$

4. Experimental Results and Analysis

In this section, the performance validation of IMDOR-MBFF method is inspected under Side-scan sonar imaging for Mine detection [30]. The dataset collects original sonar imaging captured between 2010 and 2021 employing a Teledyne Marine Gavia Autonomous Underwater Vehicle (AUV) that involves enough data for classifying its content objects as Mine-Like Contacts (MILCO) as well as Non-Mine-like Bottom Objects (NOMBO). The dataset consists of 437 instances on MILCO class and 231 instances on NOMBO class, as shown in Table 2. Fig 5 represents the class distribution. A few sample images of original and pre-processed are depicted in Fig. 6. Fig. 7 signifies a sample of original and predicted images.

Table 2 Details of dataset

Classes	Train	Test	Total
MILCO	315	122	437
NOMBO	178	53	231

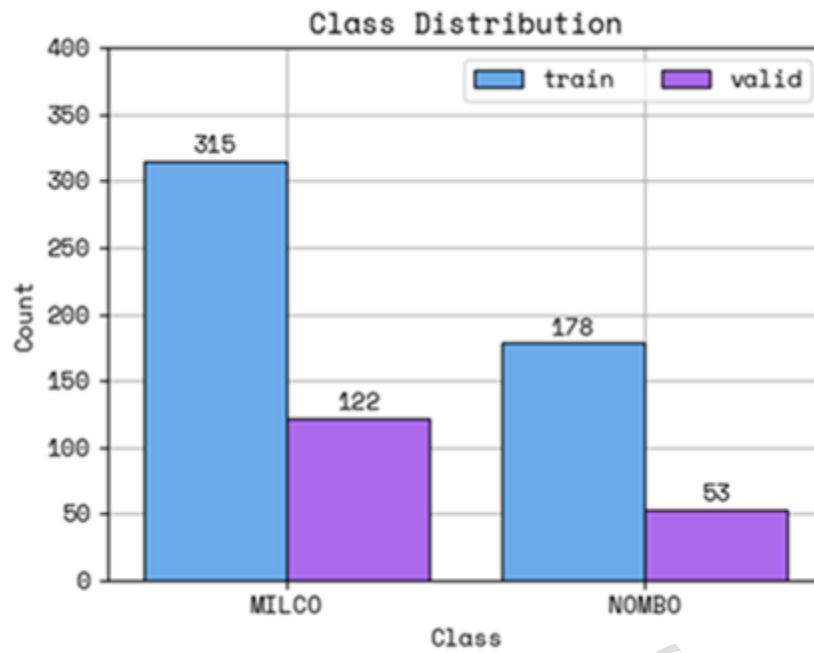


Fig. 5. Class distribution

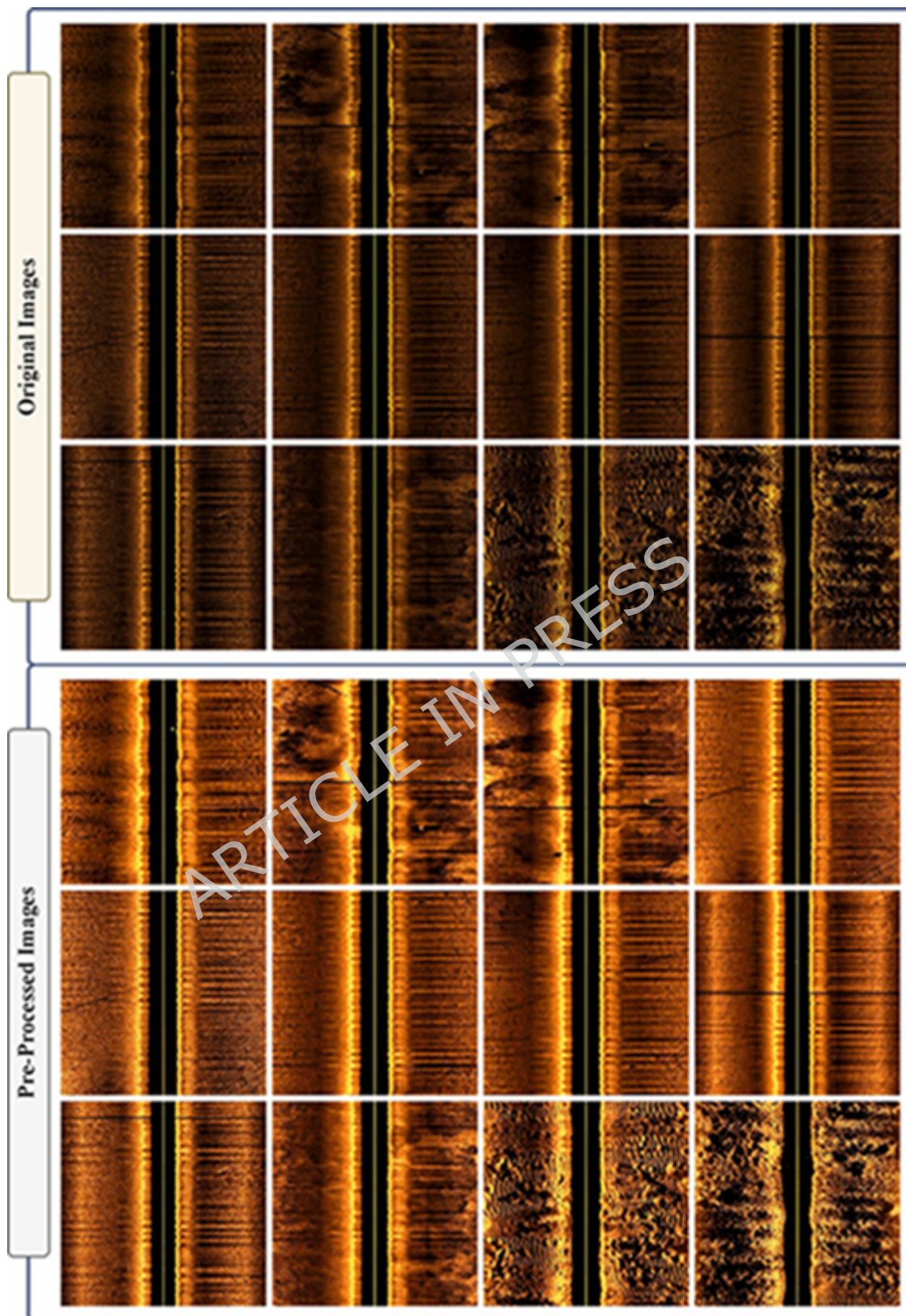


Fig. 6. Sample images of original and pre-processed

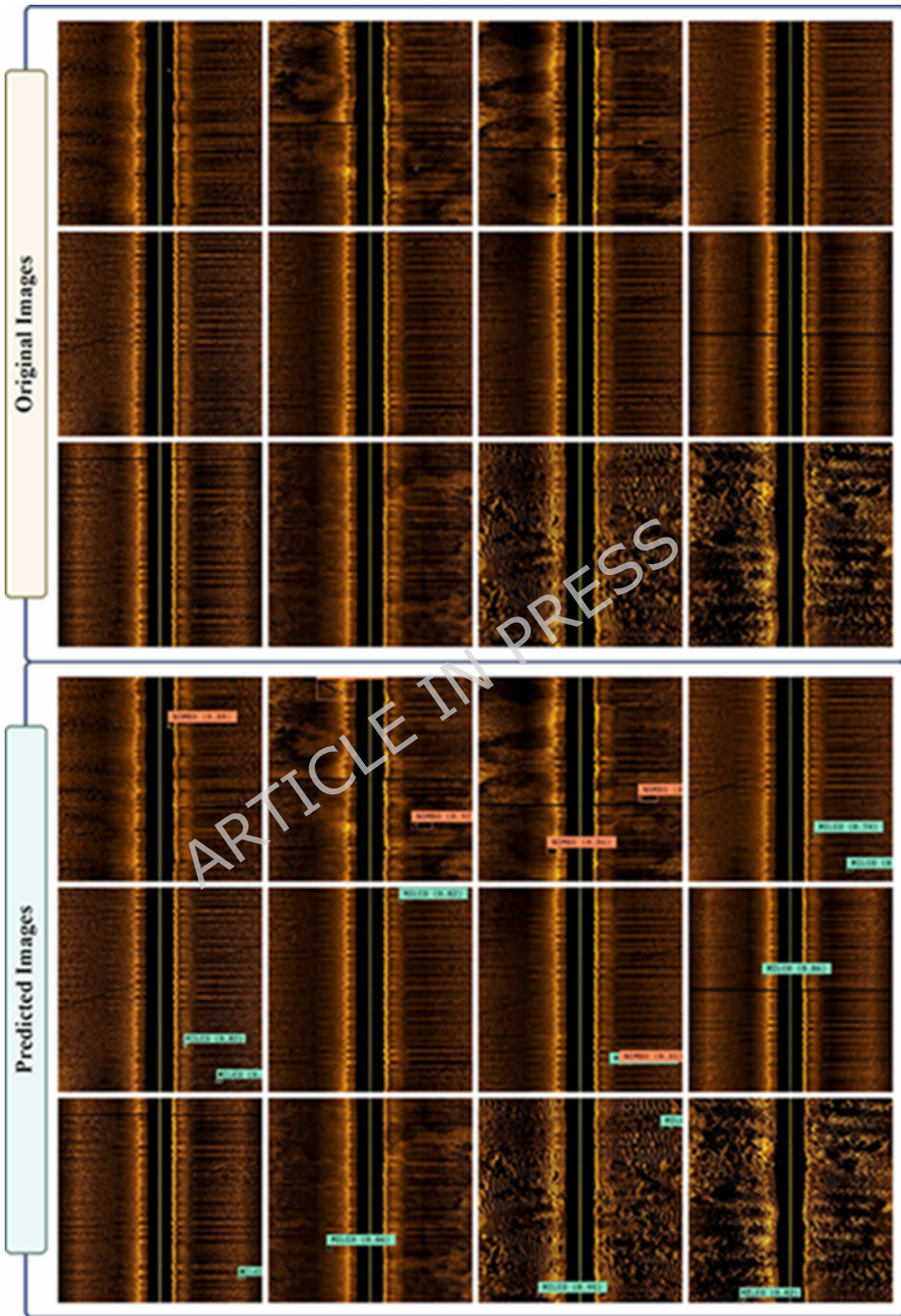


Fig. 7. Sample of original and predicted images

Table 3 and Fig. 8 show the object detection values of IMDOR-MBFF approach with training and validation. The outcome indicated that the IMDOR-MBFF model properly classified the samples. With training, the IMDOR-MBFF method provides outcomes with preci_n of 94.79%, recal_l of 88.04%, $F1_{\text{score}}$ of 91.29%, IoU of 84.75%, and mAP of 85.25%, respectively. In the meantime, with validation, preci_n of 79.83%, recal_l of 61.82%, $F1_{\text{score}}$ of 69.45%, IoU of 66.53%, and mAP of 49.89%, correspondingly.

Table 3 Object detection outcomes of IMDOR-MBFF method

Classes	Precision	Recall	F1-Score	IoU	mAP
Training					
MILCO	96.85	89.64	93.11	86.75	86.90
NOMBO	92.73	86.44	89.47	82.75	83.60
overall	94.79	88.04	91.29	84.75	85.25
Validation					
MILCO	86.49	59.81	70.72	72.59	50.68
NOMBO	73.17	63.83	68.18	60.47	49.10
overall	79.83	61.82	69.45	66.53	49.89

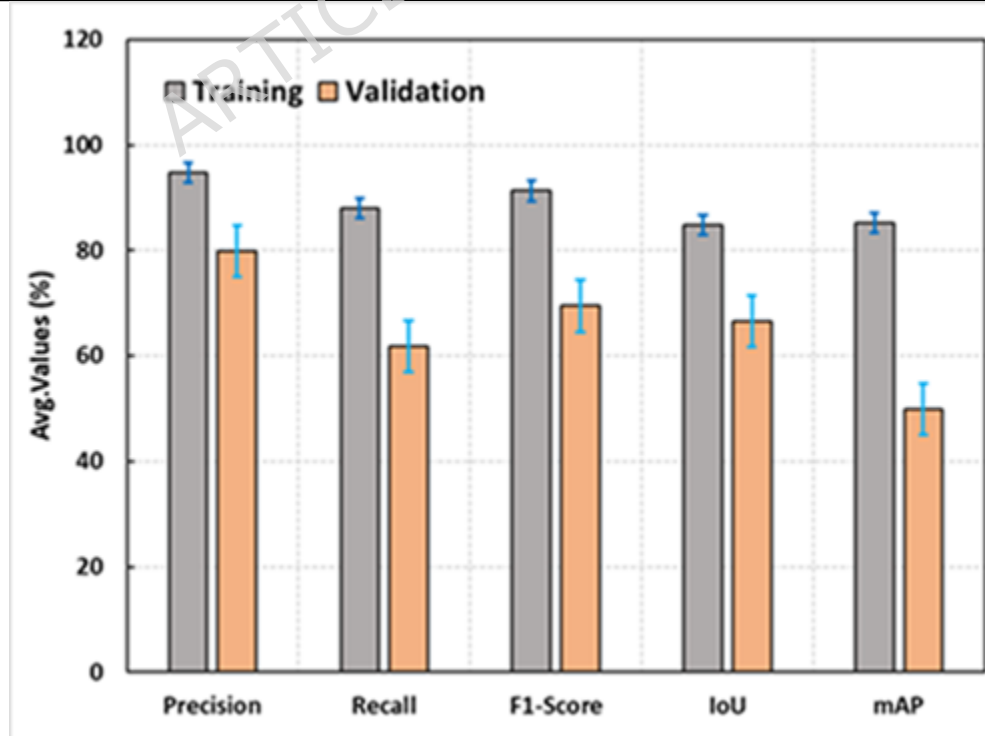


Fig. 8. Object detection outcomes of IMDOR-MBFF method

Fig. 9 represents the training and validation performance of an object detection method over epochs. Training and validation losses (box, class, and DFL) consistently decline, representing constant learning and better convergence. Precision and recall rise over time, displaying enhanced detection accuracy and fewer missed or false detections. mAP@50 and mAP@50-95 progressively increase, confirming good overall detection outcomes. Overall, the method learns efficiently with no major signs of overfitting.

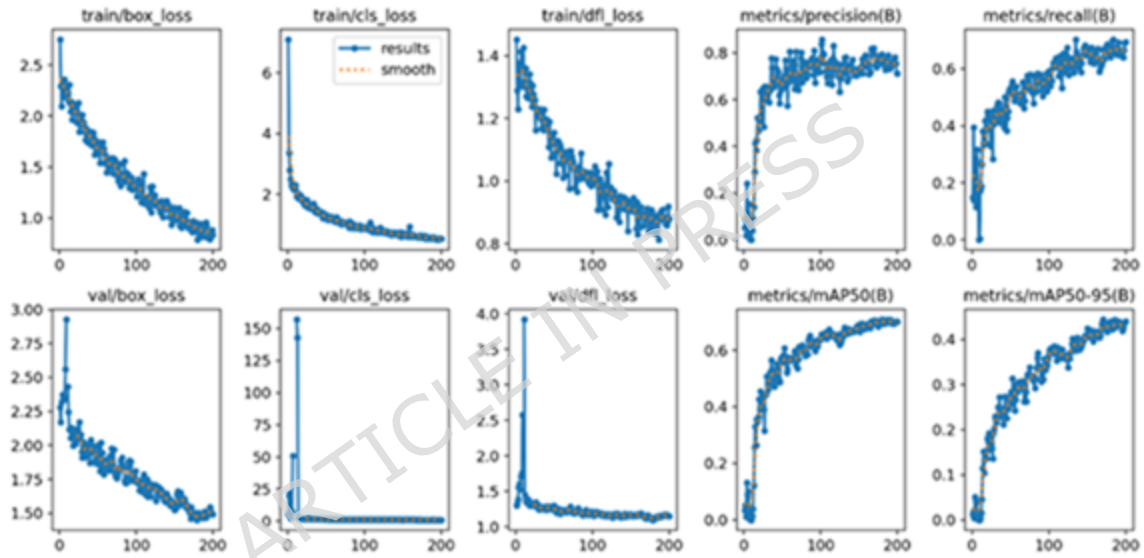
**Fig. 9.** Loss curve of IMDOR-MBFF method



Fig. 10. Training metrics of IMDOR-MBFF method

Fig. 10 proves the model's training performance over 200 epochs, exhibiting stable enhancement through each metric. Precision and recall increase suddenly in the initial phase and then slowly stabilize, signifying that the model becomes more precise and constant in object detection. The mAP@50 curve increases smoothly, reflecting enhanced quality of detection at a standard IoU threshold, while mAP@50-95 rises more gradually because of its stricter valuation criterion. Overall, the tendencies imply efficient learning and model convergence in training.

Fig. 11 depicts a comparative analysis between two groups, MILCO and NOMBO. Subsequently, the bar plot displays that MILCO has a huge number of samples related to NOMBO. Then, the panel shows variability in the data, with MILCO showing a broad spread, implying more variation or diversity within the group. In addition, the scatter plot exhibits two different clusters along the x-axis, likely consistent with the dual groups, revealing differences in their spatial characteristics or distribution. Lastly, the plots exemplify that both groups predominantly contains sample with relatively smaller width and

height values, demonstrating similarity in size-related features despite other variances.

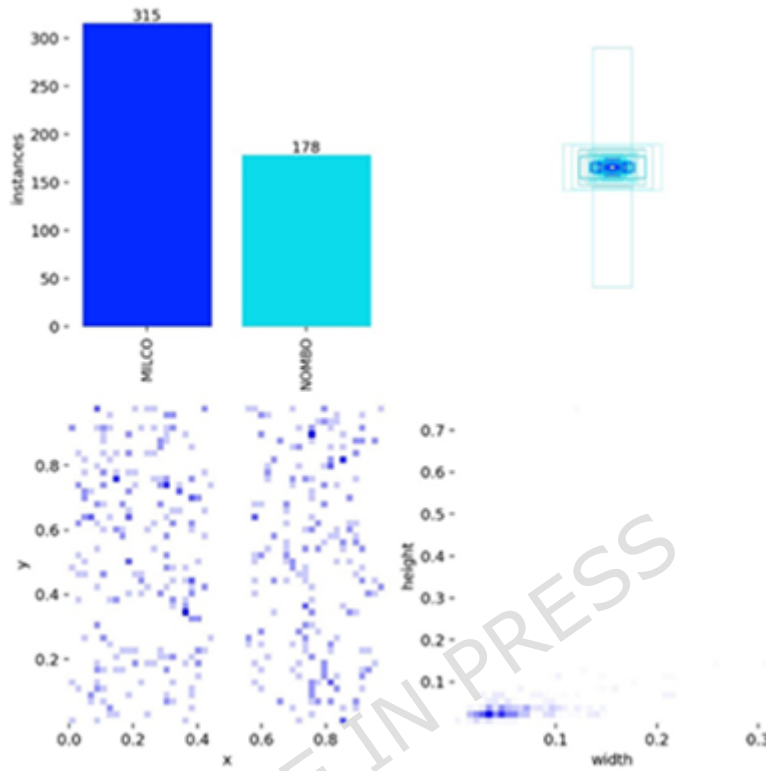


Fig. 11. Mine detection dataset, anchor boxes, x and y coordinates, width and height coordinates

Fig. 12 represents a series of sonar scans with detected objects labeled as either MILCO or NOMBO, every achieves a confidence score signifying the model's certainty in the classification. The labels are color-coded blue for MILCO and cyan for NOMBO and display numerous detections per scan, reflecting the model's capability of identifying and differentiating among these dual categories within intricate underwater images.



Fig. 12. Prediction outcome of IMDOR-MBFF method

Table 4 and Fig. 13 indicate the comparative outcomes of IMDOR-MBFF methods with existing models. The experimental outcome inferred that the IMDOR-MBFF model has shown better performance. Fig. 13 (a) denotes the comparative preci_n and recal_l outcomes with existing methodologies. Based on preci_n , the outcome demonstrated that the Refine Det model gives the worst performance with preci_n of 68.66%, whereas the Cascaded Way, Underwater YCC, Parallel Way, and BG-YOLO techniques got slightly greater outcomes with preci_n of 69.15%, 76.28%, 70.45%, and 70.33%, correspondingly. Moreover, the SSD512 and YOLOv5s techniques provide considerable performance with preci_n of 80.53% and 81.69%. Additionally,

the proposed IMDOR-MBFF method attains maximal performance with prec_n of 94.79%. Along with that, based on recal_l , the result exhibited that the SSD512 model got ineffective performance with recal_l of 63.55%, whereas the Refine Det, Parallel Way, and BG-YOLO approaches got moderately higher performance with recal_l of 66.94%, 71.55%, and 70.15%, respectively. Besides, the YOLOv5s, Underwater YCC, and Cascaded Way systems attain reasonable outcomes with recal_l of 74.62%, 78.75%, and 82.80%, correspondingly. Eventually, the proposed IMDOR-MBFF method accomplishes superior outcomes with recal_l of 88.04%.

Table 4 Comparison of outcomes of IMDOR-MBFF approach with existing models

Methods	Precision	Recall	IoU	mAP
SSD512	80.53	63.55	65.41	80.04
Refine Det	68.66	66.94	79.23	70.19
YOLOv5s	81.69	74.62	70.31	72.89
Underwater YCC	76.28	78.75	65.62	81.84
Cascaded Way	69.15	82.80	77.92	68.58
Parallel Way	70.45	71.55	75.31	77.74
BG-YOLO	70.33	70.15	80.23	62.22
IMDOR-MBFF	94.79	88.04	84.75	85.25

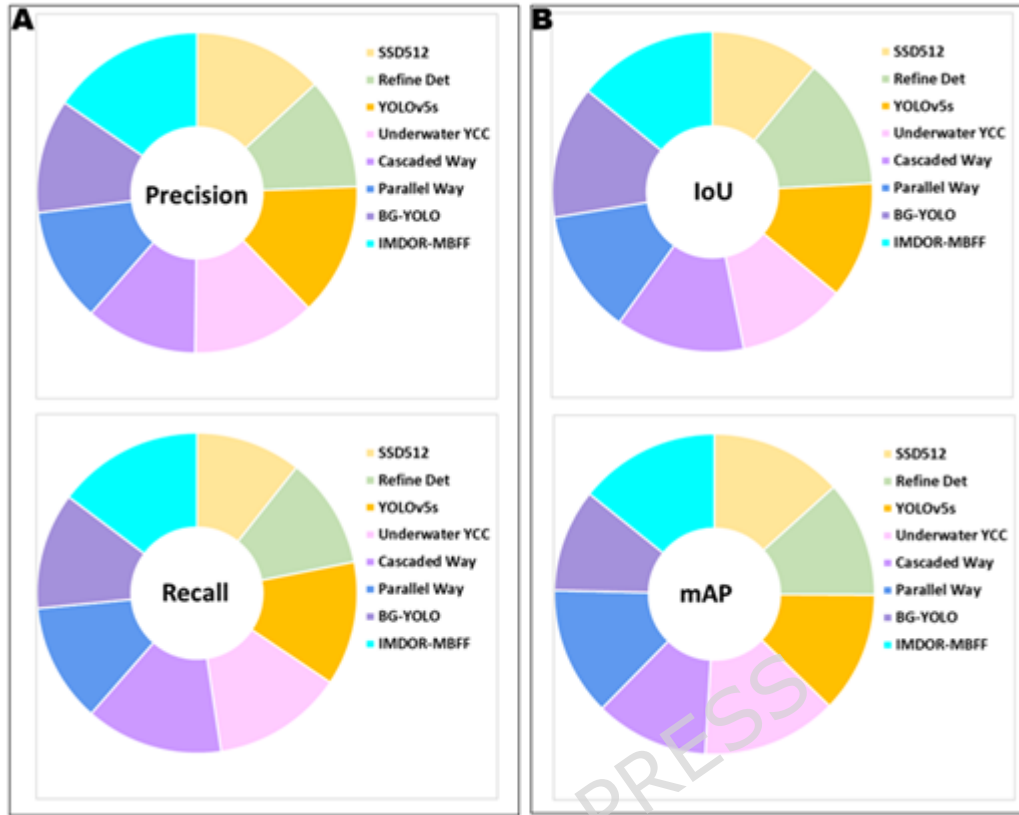


Fig. 13. Comparative outcomes of IMDOR-MBFF method (a) $prec_i$ and $recal_i$ (b) IoU and mAP

Fig. 13 (b) experimental IoU and mAP outcomes of IMDOR-MBFF method with existing methodologies. Based on IoU, the SSD512 and Underwater YCC systems offer ineffective results, whereas the YOLOv5s, Cascaded Way, and Parallel Way techniques attain slightly higher solutions. Moreover, the Refine Det and BG-YOLO methods gain somewhat higher outcomes. Likewise, the proposed IMDOR-MBFF method accomplished maximal performance with IoU of 84.75%. In the meantime, based on mAP, the proposed IMDOR-MBFF method attains superior performance with mAP of 85.25%, whereas the SSD512 and Underwater YCC systems got slightly lower outcomes. Also, the Refine Det, YOLOv5s, and Parallel Way approaches achieve somewhat lower performance. Ultimately, the Cascaded Way and BG-YOLO systems had ineffective performance.

Fig. 14 proves the confusion matrices created by the IMDOR-MBFF model under TRAP/TESP. The result suggests that the IMDOR-MBFF approach has effective classification and detection of each class exactly.

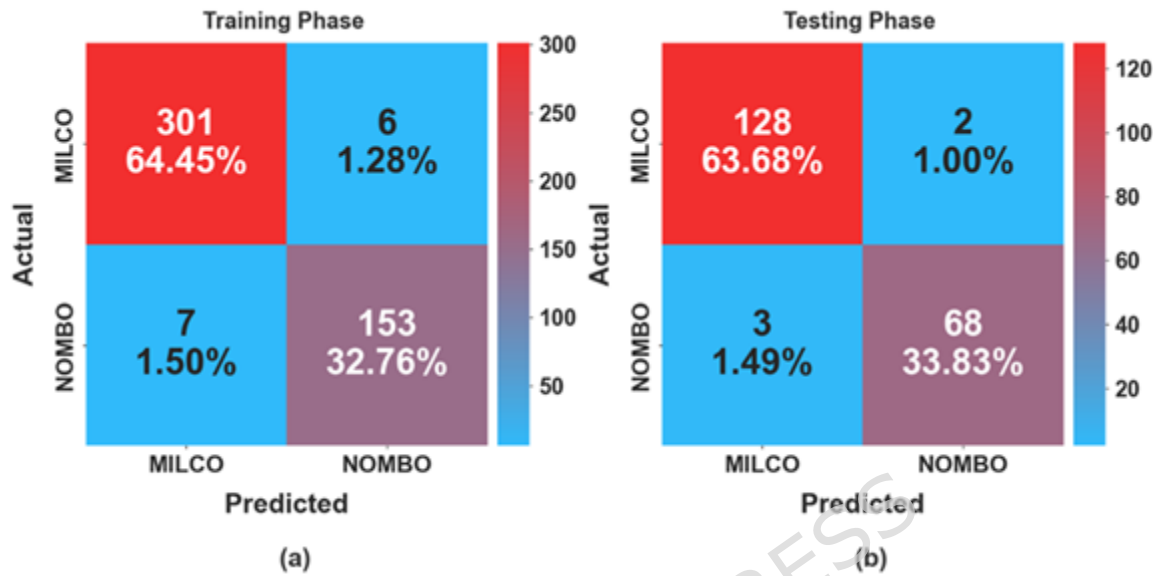


Fig. 14. Confusion matrices of IMDOR-MBFF method (a) TRAP and (b) TESP

Table 5 and Fig. 15 depict the classifier outcomes of IMDOR-MBFF method with TRAP and TESP. The outcome indicates that the IMDOR-MBFF model properly classified the samples. With TRAP, the IMDOR-MBFF model gives average with acc_{ur_y} of 96.84%, $prec_{i_n}$ of 96.98%, $recal_l$ of 96.84%, $F1_{score}$ of 96.91%, and MCC of 93.81%, respectively. At the same time, with TESP, the IMDOR-MBFF method provides average with acc_{ur_y} of 97.12%, $prec_{i_n}$ of 97.43%, $recal_l$ of 97.12%, $F1_{score}$ of 97.27%, and MCC of 94.54%, correspondingly.

Table 5 Classifier outcomes of IMDOR-MBFF method with TRAP and TESP

Classes	Accuracy	Precision	Recall	F1-Score	MCC
Training Phase					
MILCO	98.05	97.73	98.05	97.89	93.81
NOMBO	95.63	96.23	95.63	95.92	93.81
Average	96.84	96.98	96.84	96.91	93.81

Testing Phase					
MILCO	98.46	97.71	98.46	98.08	94.54
NOMBO	95.77	97.14	95.77	96.45	94.54
Average	97.12	97.43	97.12	97.27	94.54

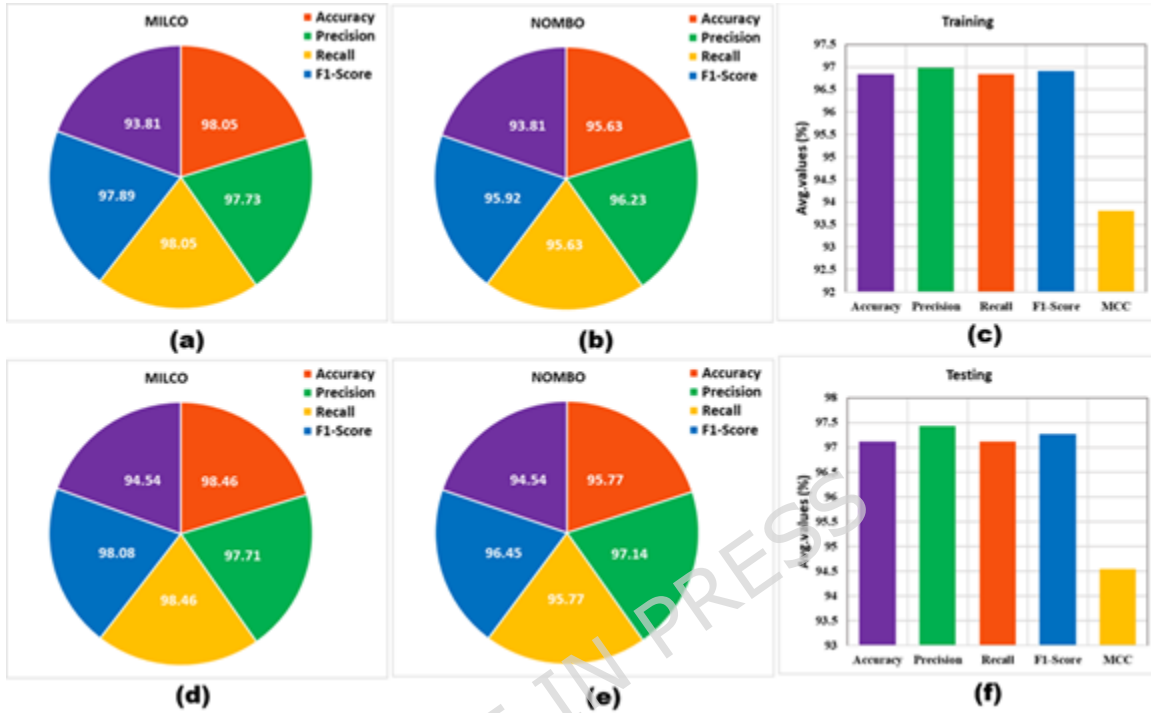


Fig. 15. Classifier outcomes of IMDOR-MBFF method with TRAP and TESP

Fig. 16 presents the training (TRAIN) and validation (VALID) accuracy curves of the IMDOR-MBFF model on TRAP and TESP over 200 epochs. TRAIN accuracy grown steadily and achieved a higher level, representing efficient learning and convergence. The VALID accuracy closely follows the training tendency with a smaller and consistent gap, signifying better generalizes and lower overfitting. Overall, the reliable improvement and convergence of both curves confirm the reliability and robustness of the proposed system.

Next, Fig. 17 portrays the TRAIN and VALID loss curves of the IMDOR-MBFF model TRAP and TESP across 200 epochs. Both losses reduce sharply in the early epochs, implying fast learning, and then slowly decrease in a steady and smooth manner, reflecting efficient convergence. The close alignment among training and validation loss, with only a minimal stable gap, implies inferior

overfitting and robust generalization ability. Overall, the stable reduction in loss confirms the constancy and effectiveness of the training procedure.

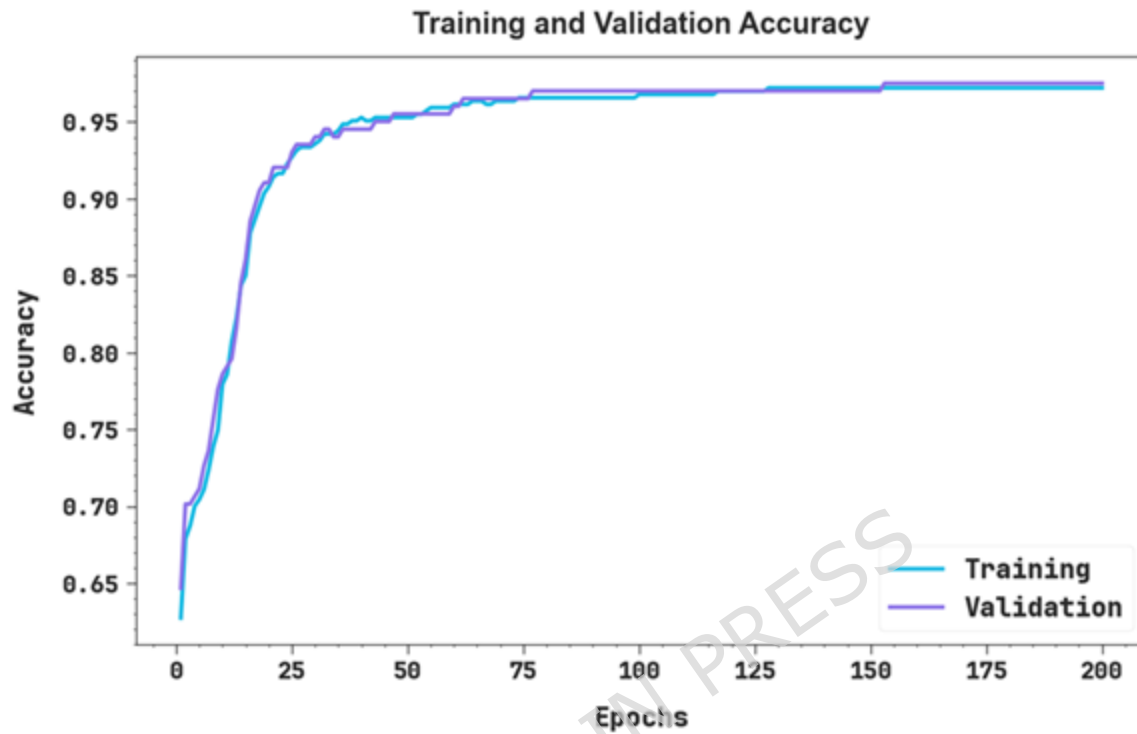


Fig. 16. Accuracy curve of IMDOR-MBFF method

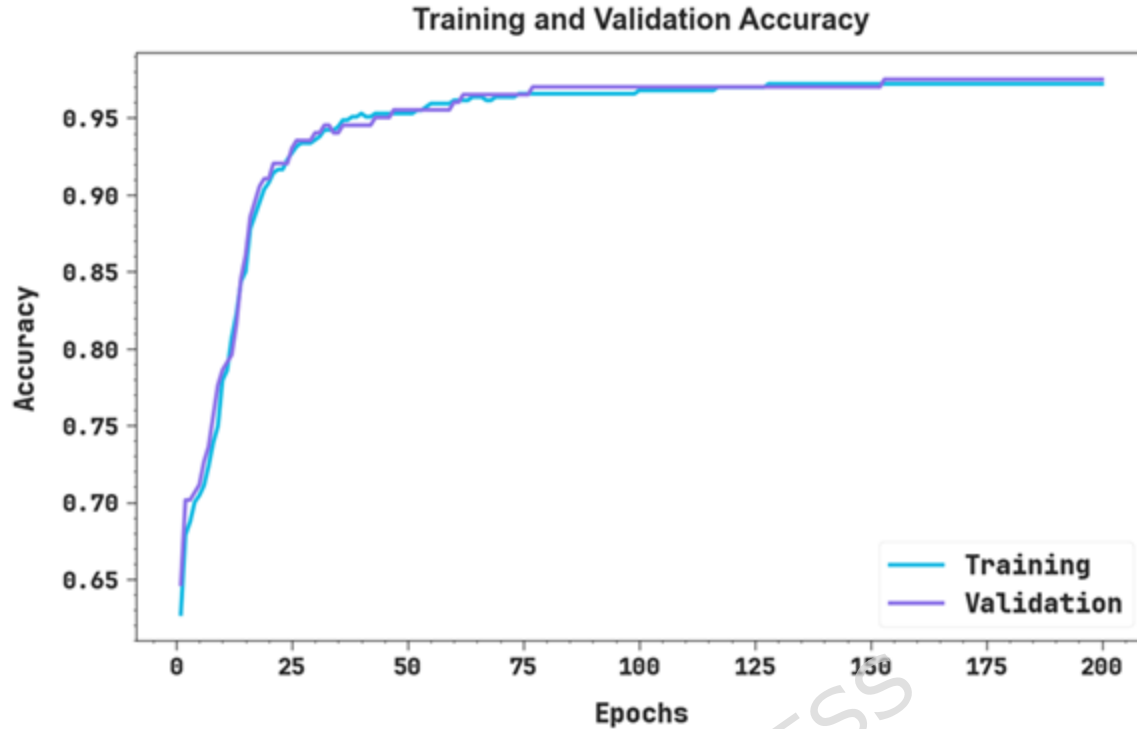


Fig. 17. Loss curve of IMDOR-MBFF method

Table 6 reveals the experimental outcomes and frame per second (FPS) of IMDOR-MBFF technique with existing models [31-36]. The simulation outcome stated that the IMDOR-MBFF method outperformed other methods. Depending upon FPS, the proposed IMDOR-MBFF approach obtains lesser outcome with FPS of 1.55sec, whereas Aqua-DETR, Boosting R-CNN, YOLOv3 237e, Auto Assign, Light-ResNet, ResNet_0, YOLOv3-tiny, and Faster-RCNN systems got somewhat superior FPS of 8.03sec, 7.22sec, 10.23sec, 9.78sec, 2.86sec, 5.20sec, 5.21sec, and 5.23sec, respectively. At last, the UNet Model, VGGNet, and YOLOv8n models got higher FPS of 7.49sec, 8.23sec, and 6.66sec, correspondingly.

Table 6 Comparative outcomes of IMDOR-MBFF method with existing models

Model	Accurac y	Precisio n	Recal l	F1-Score	Frame Second (FPS)	Per
Aqua-DETR	86.90	82.80	61.35	68.61	8.03	

Boosting R-CNN	86.13	79.60	78.67	81.90	7.22
YOLOv3 237e	44.87	37.80	34.90	41.93	10.23
Auto Assign	40.32	35.50	35.45	36.27	9.78
ResNet_0	88.10	86.74	90.15	89.18	5.20
Light-ResNet	86.10	88.21	89.97	90.96	2.86
UNet Model	86.00	86.74	83.47	87.63	7.49
VGGNet	94.50	90.47	87.10	87.09	8.23
YOLOv3-tiny	95.33	95.78	92.25	88.45	5.21
YOLOv8n	90.34	96.25	87.30	89.39	6.66
Faster-RCNN	94.11	91.13	92.99	94.86	5.23
IMDOR-MBFF	97.12	97.43	97.12	97.27	1.55

Fig. 18 shows the comparison results of the proposed model. Fig. 18 (a) signifies the comparative $accu_y$ outcomes of IMDOR-MBFF method with existing systems. The outcome exhibited that the proposed IMDOR-MBFF method attains superior performance with $accu_y$ of 97.12%, whereas the Aqua-DETR , VGGNet, YOLOv3-tiny, and Faster-RCNN approaches gives slightly lower performance with $accu_y$ of 86.90%, 94.50%, 95.33%, and 94.11%, respectively. Moreover, the YOLOv8n and ResNet_0 techniques got considerable outcomes. Likewise, the Light-ResNet and UNet models got minimal performance. Fig. 18 (b) displays the comparison $prec_i_n$ results of IMDOR-MBFF method with existing systems. The result exhibited that the YOLOv3 237e, ResNet_0 and UNet techniques got minimal outcomes with $prec_i_n$ of 37.80%, 86.74% and 86.74%, whereas the Boosting R-CNN , Auto Assign , Light-ResNet and VGGNet approaches offers slightly greater outcomes with $prec_i_n$ of 79.60%, 35.50%, 88.21% and 90.47%. Furthermore, the YOLOv3-tiny, YOLOv8n, and Faster-RCNN techniques attains somewhat closer outcomes with $prec_i_n$ of 95.78%, 96.25%, and 91.13%, correspondingly. Finally, the proposed IMDOR-MBFF method accomplishes superior result with $prec_i_n$ of 97.43%.

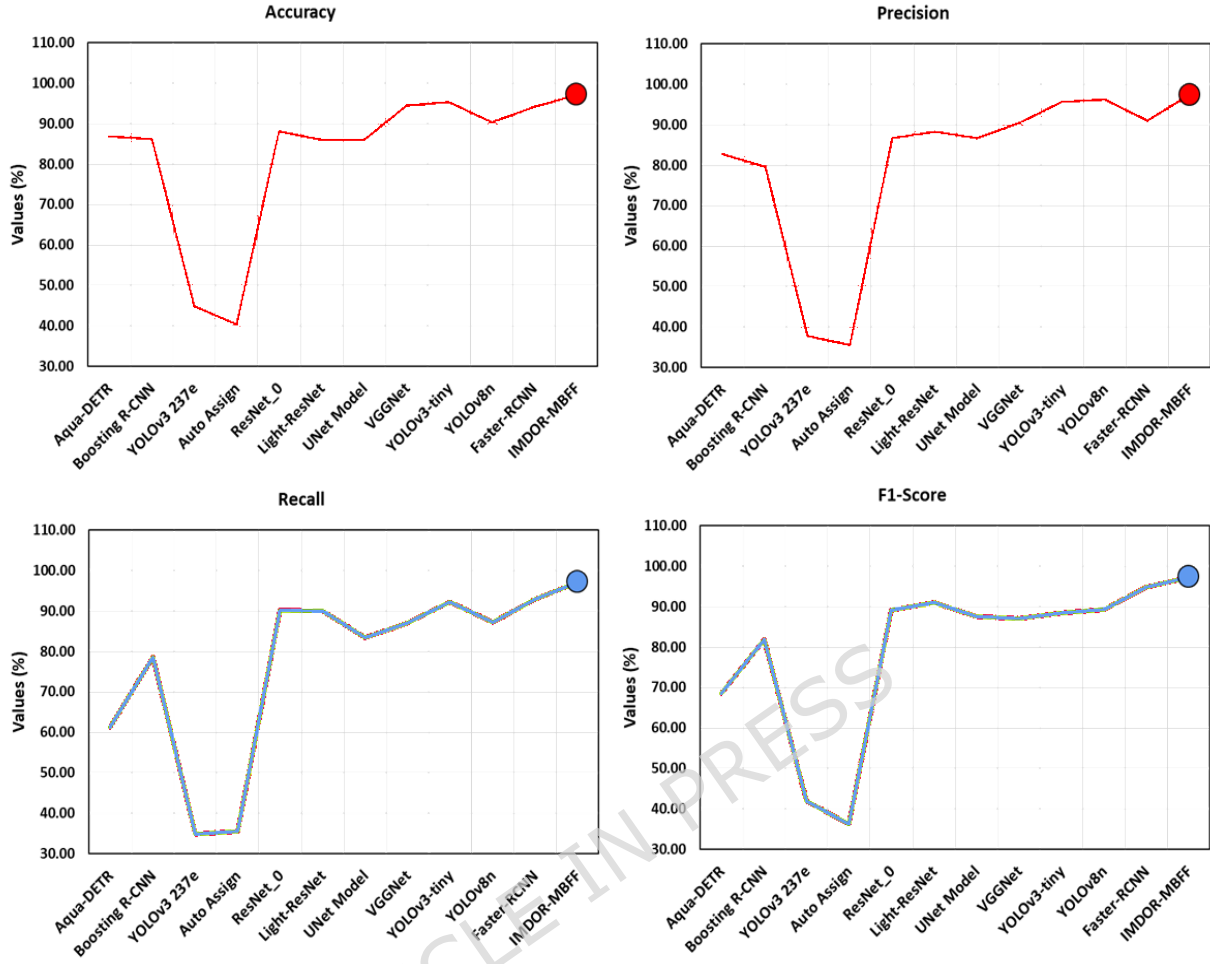


Fig. 18. Comparative outcomes of IMDOR-MBFF method (a) $accu_r$, (b) $preci_n$, (c) $recal_l$, and (d) $F1_{score}$

Fig. 18 (c) comparative $recal_l$ analysis of IMDOR-MBFF method with existing models. The UNet method shows worst performance. Besides, the VGGNet, YOLOv8n, and Light-ResNet systems attains slightly higher results. Along with that, the ResNet_0, YOLOv3-tiny and Faster-RCNN techniques got closer performance. Eventually, the proposed IMDOR-MBFF method attains greater performance with $recal_l$ of 97.12%, when compared to other models.

Table 7 reveals the computational efficiency of the IMDOR-MBFF method with existing approaches. The proposed IMDOR-MBFF method exhibits lower computational efficiency with FLOPs of 5.23G, GPU of 876M, and inference time of 0.87ms, correspondingly.

Table 7 Computational efficiency of IMDOR-MBFF method with existing approaches

Model	FLOPs (G)	GPU (M)	Inference Time (ms)
ResNet_0	91.60	4395	5.50
Light-ResNet	27.30	4079	2.31
UNet	112.50	2673	4.91
VGGNet	130.10	1775	4.35
YOLOv6	11.80	3238	2.11
RT-DETR	37.60	4863	4.75
IMDOR-MBFF	5.23	876	0.87

At the same time, the ResNet_0, Light-ResNet, UNet, VGGNet, YOLOv6, and RT-DETR methodologies records highest computational efficiency with FLOPs of 91.60G, 27.30G, 112.50G, 130.10G, 11.80G, and 37.60G, respectively. Consequently, the presented IMDOR-MBFF method is effectual one for mine detection and object recognition using sonar images that highlights superior performance.

Table 8 presents the K-fold cross validation results obtained by the proposed model. The table values demonstrated that the proposed model attains effectual performance under each fold in terms of different measures.

Table 8 K-Fold Cross Validation of the IMDOR-MBFF Method

K-Fold Cross Validation	Accuracy	Precision	Recall	F1-Score
Fold-1	93.78	94.77	90.34	95.28
Fold-2	94.51	93.71	95.00	93.76
Fold-3	93.98	93.65	91.74	90.95
Fold-4	91.98	90.25	89.37	89.23
Fold-5	92.34	91.38	91.66	94.89
Fold-6	95.80	92.34	94.35	94.66
Fold-7	92.01	92.11	95.33	91.90
Fold-8	92.17	94.22	93.19	89.00
Fold-9	93.16	95.17	90.05	90.33

Fold-10	94.84	90.04	90.72	90.55
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Table 9 shows the ablation study investigation of the IMDOR-MBFF algorithm. The IMDOR-MBFF model gains highest $accu_y$ of 97.12%, $prec_n$ of 97.43%, $reca_l$ of 97.12%, and $F1_{Score}$ of 97.27%, creating the ability of IMDOR-MBFF (Temmporal attention using Attention Mechanism) approaches, while RetinaNet (using objecti detection) presented an $accu_y$ of 93.32%, $prec_n$ of 93.25%, $reca_l$ of 93.07%, and $F1_{Score}$ of 93.44%. Similarly, GhostNetV2 only (feature extraction) better metrics to an $accu_y$ of 93.86%, $prec_n$ of 93.99%, $reca_l$ of 93.83%, and $F1_{Score}$ of 94.23%. Moreover, ConvNeXt only (feature extraction) attained an $accu_y$ of 94.36%, $prec_n$ of 94.75%, $reca_l$ of 94.40%, and $F1_{Score}$ of 94.80%. Also, ShuffleNetV1 only (feature extraction) realized an $accu_y$ of 95.02%, $prec_n$ of 95.50%, $reca_l$ of 95.12%, and $F1_{Score}$ of 95.45%, indicating further improvements. Likewise, ShuffleNetV1 only (feature extraction) algorithms achieved an $accu_y$ of 95.61%, $prec_n$ of 96.11%, $reca_l$ of 95.86%, and $F1_{Score}$ of 96.04%. Furthermore, Triple Backbone Fusion (GhostNetV2 + ConvNeXt + ShuffleNetV1 without Temporal Unit) reached an $accu_y$ of 96.32%, $prec_n$ of 96.86%, $reca_l$ of 96.51%, and $F1_{Score}$ of 96.59%, importance slightly better performance.

Table 9 Ablation of the IMDOR-MBFF algorithms

Model	Accuracy	Precision	Recall	F1-Score
RetinaNet (using objecti detection)	93.32	93.25	93.07	93.44
GhostNetV2 only (feature extraction)	93.86	93.99	93.83	94.23
ConvNeXt only (feature extraction)	94.36	94.75	94.40	94.80
ShuffleNetV1 only (feature extraction)	95.02	95.50	95.12	95.45

Dual Backbone Fusion (Two feature extractors only)	95.61	96.11	95.86	96.04
Triple Backbone Fusion (GhostNetV2 + ConvNeXt + ShuffleNetV1 without Temporal Unit)	96.32	96.86	96.51	96.59
IMDOR-MBFF (Temporal attention using Attention Mechanism)	97.12	97.43	97.12	97.27

5. Conclusion

In this manuscript, we have concentrated on the design of the IMDOR-MBFF approach in side-scan sonar imaging data. The major objective of the IMDOR-MBFF framework is to efficiently enhance detection and classification performance for underwater mine detection using an AUV. Primarily, a gamma distribution-based speckle noise method with logarithmic transformation has been used as the image pre-processing phase to suppress multiplicative noise and improve structural details. Subsequently, object detection has been accomplished using a RetinaNet framework with backbone fusion to accurately localize mine-like targets under complex seabed conditions. Furthermore, discriminative features are extracted through a lightweight yet powerful fusion of the GhostNetV2 model, ConvNeXt method, and ShuffleNetV1 technique is capturing both fine-grained textures and global contextual data. After the feature extractor, a feature fusion approach that depends on concatenation has been utilized to efficiently integrate the complementary data extracted from the three backbones. Lastly, a TAGU was deployed for precise mine classification by modeling temporal dependencies across sonar image sequences. The experimental performance of the proposed IMDOR-MBFF framework is assessed on a publicly available benchmark side-scan sonar imaging dataset for mine detection sourced from the Kaggle repository. The comparative analysis demonstrates its superior performance over existing methodologies across multiple evaluation metrics.

Though the proposed model accomplishes better performance, it suffers from few limitations. presented methodology works on simple channel concatenation for multi-backbone fusion without discovering additional progressive fusion strategies, and it doesn't clearly address class imbalance in the database, which may influence simplification and unfairness the performance. Future work will examine attention-based or adaptive fusion mechanisms to increase feature combination and incorporate imbalance-handling techniques such as focal loss, class reweighting, and data augmentation to improve robustness and consistency.

Data Availability Statement

The data that support the findings of this study are openly available in the Kaggle repository at <https://www.kaggle.com/datasets/sierra022/sonar-imaging-mine-detection/data>, reference number [29].

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